

CURRICULUM VITAE

PARK GYU HYEON

EMBEDDED SYSTEMS & ROBOTICS CONTROL ENGINEER

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Electrical and Electronic Engineering student focused on embedded systems and robotics control. Experienced in building complete robotic systems spanning custom PCBs, mechanical design, real-time MCU firmware, perception, localization, and ROS integration. Proven through national competition wins, an international top-10 finish, and selection as a Presidential Science Scholar.

EDUCATION

Dankook University

Mar 2022 - Expected Feb 2028

B.S. IN ELECTRICAL AND ELECTRONIC ENGINEERING

GPA 4.27/4.50 97.7/100 93 credits completed

SCHOLARSHIP

Presidential Science Scholar

3rd-Year National Selection **2026 - 2027**

COMPETITION ROBOTICS

Domestic High-Speed Line Tracer

2025

INDEPENDENT DEVELOPER

GitHub: Linetracer

C, Assembly, TI TMS320F2809, 2 kHz Control, PCB Design, SolidWorks, EasyEDA

- Designed the complete platform across mechanics, sensor circuitry, custom PCBs, firmware, and segment-specific motion profiles.
- Implemented a 2 kHz loop with 25 us sensor processing, pulsed emission, ambient-light filtering, ADC optimization, and time-bounded control execution.
- Won Grand Prize at two national competitions, with official records of 15.921 s and 25.387 s.

2 kHz	15.921 s	2	Full-stack
control loop	competition record	national grand prizes	robot design

All Japan Robotrace Platform - SANDEVISTAN

2025 - Feb 2026

INDEPENDENT DEVELOPER

GitHub: JP_Robotrace

C, Assembly, TI TMS320F2809, Integrated PCB Chassis, Suction System

- Re-engineered the platform for tighter international track geometry using a 163 g integrated-PCB chassis and suction-fan system.
- Recovered a damaged suction system immediately before competition and finished 8th with a 13.074 s official record.

Multi-Robot Disaster Search and Rescue System

Jul 2026 - Present

DEVELOPER

GitHub: damgc_robot

ROS 2, C++, Python, STM32, RealSense D435, YOLO11n, AprilTag, TF2, EKF, VSLAM

- Developing leader/follower robots for survivor search and cooperative relief-supply transport.
- Built RGB-D survivor localization and guarded velocity pipelines integrated with STM32 motor, encoder, IMU, PID, odometry, and watchdog components.

2025 Microcontroller Application Contest

Nov 2025

TEAM LEADER - 4 MEMBERS

GitHub: 2025-COSS-MCU-Contest

STM32, AVR, ATtiny85, HUB75 LED Matrix, Interrupt-Driven Firmware

- Led architecture and integration; implemented interrupt-driven PWM and a compact 3-bit RGB buffer architecture for stable output under limited memory.
- Achieved 1st place and the Minister of Education Award.

PARK GYU HYEON

Robotics & Autonomous Systems

github.com/rbgusrns

AUTONOMOUS DRIVING

Gyeonggi Autonomous Driving Challenge 2026

2025 - Feb 2026

PERCEPTION & CONTROL DEVELOPER

GitHub: Gyeonggi-AD-Challenge-2026

Python, PyTorch, YOLO, OpenCV, BEV, RPLidar

- Replaced unreliable end-to-end imitation learning with AI-based lane perception and deterministic control to handle mechanical steering offsets.
- Implemented lane segmentation, BEV transformation, traffic-light recognition, and LiDAR obstacle avoidance; received the Encouragement Prize.

Kookmin Self-Driving Competition

2025

AUTONOMOUS DRIVING DEVELOPER

GitHub: Kookmin_Self_Driving

Python, ROS, OpenCV, Xycar, Camera, LiDAR

- Integrated HSV/BEV and Hough lane perception, green-light start logic, obstacle navigation, and lane-change behavior.

AMET Autonomous Driving Hackathon

2025

SOFTWARE DEVELOPER

GitHub: AMET

Python, ROS, OpenCV, Xycar, LaserScan

- Built sliding-window lane following, PD steering, traffic-light detection, and LiDAR-triggered overtaking and lane-return behavior.

SERVICE & INTERACTIVE ROBOTICS

MediWay Hospital Assistance Robot

2025

INDOOR LOCALIZATION DEVELOPER

GitHub: Robot-with-Beacon-Navigator

ROS Noetic, C++, Python, LiDAR, IMU, EKF

- Contributed to a hospital navigation and medicine-delivery robot integrating localization, route guidance, and autonomous driving.

BLE Indoor Localization Engine

2025

LOCALIZATION DEVELOPER

GitHub: indoor-localization

Python, BLE Fingerprinting, LightGBM, A*, PyQt

- Implemented beacon scanning, fingerprint databases, learned position prediction, heatmaps, path planning, and real-time map visualization.

Autonomous Air Hockey Robot

2023 - 2024

MOTOR CONTROL DEVELOPER

GitHub: Air-hockey

C, Python, TI TMS320F2808, OpenCV, SCI

- Connected puck tracking and trajectory prediction with a two-axis embedded striker; received a 2024 ICT project award.

Wearable-Controlled Collaborative Robot Arm

2026

TEAM LEAD & EMBEDDED DEVELOPER

GitHub: Waerable-Robot_ARM

STM32G474, FreeRTOS, C, Dynamixel

- Led the team and developed task-based STM32 firmware and Dynamixel motor interfaces for a wearable-input collaborative robot arm.

JOG MATE - Assistive Jogging Robot

2025

HARDWARE FABRICATION

Mobile Robotics, Fabrication, Obstacle Detection, Human-Centered Design

- Built hardware for a track-following assistive robot providing pacing feedback and obstacle awareness for visually impaired runners.
- Received an ECHO+ Excellence Award and an Intelligent Robotics Consortium Creative Design Award.

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Embedded Platforms, Leadership & Recognition

github.com/rbgusrns

EMBEDDED & SOFTWARE PROJECTS

STM32G474 Robotics Education Board

GitHub: M4-Common-board

Designed the MAZE education platform with modular ADC, motor, sensor, OLED, flash, SPI, I2C, and UART firmware.

STM32G474 - C - CMake

STM32H533 Next-Generation Robotrace

GitHub: M33_26JP

Personal Cortex-M33 platform with route logging, map correction, profile optimization, external flash, gyro, odometry, and race-control modules.

STM32H533 - C - OctoSPI

Smart Aquaculture System

GitHub: Yangsikjang

Built a dual-Raspberry Pi system with a touch UI, MQTT communication, real-time sensing, manual control, and rule-based automation.

Python - PyQt5 - MQTT - Raspberry Pi

Data Structures & Algorithms in C

GitHub: Data-Structures-Algorithms-With-C

Implemented linked structures, queues, stacks, trees, heaps, hashing, sorting, and topological sort exercises.

C - Algorithms - Data Structures

Linux Systems Programming Study

GitHub: LinuxSTudy

Practiced modular C builds, Makefiles, object files, low-level compilation outputs, and command-line development.

Linux - C - Make

LEADERSHIP

President, MAZE Robotics Club

Sep 2025 - Sep 2026

DANKOOK UNIVERSITY

- Replaced the TI TMS320F2809 education platform with an STM32G474 ARM Cortex-M4 system and a VS Code workflow.
- Created updated theory and hands-on firmware materials.
- Introduced stencil, reflow, and hot-air processes, improving PCB assembly yield from about 30% to 90% across 30 builds.
- Coordinated competitions, procurement, and knowledge transfer.

HONORS & AWARDS

Presidential Science Scholar — 2026 - 2027
3rd-Year National Selection

Minister of Education Award — Dec 2025
MCU Application Contest

8th Place — Feb 2026
All Japan Micromouse - Robotrace

Grand Prize — Aug 2025
National Intelligent Robot Competition

Grand Prize — Aug 2025
National Line Tracer Competition

Excellence Award — Aug 2025
Robot Convergence Festival

Bronze Prize — Nov 2025
Hanium ICT Mentoring Competition

Additional Team Awards — 2024 - 2026
ECHO+, Robotics Consortium, Gyeonggi AD, ICT Convergence

TECHNICAL SKILLS

EMBEDDED SYSTEMS
STM32G474/H533, TI C2000, AVR, ATtiny85, interrupts, ADC, PWM, DMA, UART, I2C, SPI, motor control

ROBOTICS & CONTROL
PID, ROS, ROS 2, TF2, EKF, VSLAM, LiDAR, AprilTag, odometry, path planning

PERCEPTION & AI
OpenCV, PyTorch, YOLO, RGB-D, BLE fingerprinting, LightGBM

PROGRAMMING
C, C++, Python, Assembly

HARDWARE & TOOLS
PCB design, board bring-up, reflow, SolidWorks, EasyEDA, Linux, Git, Docker, CMake, VS Code

MILITARY SERVICE

Republic of Korea Army - Sergeant — Apr 2023 - Oct 2024